

Similarly, people tend to perceive probabilities and odds that resonate with their own preexisting ideas—even when the resulting conclusions drawn are statistically invalid. For example, the “Gambler’s Fallacy” refers to the commonly held impression that gambling luck runs in streaks. However, subjective psychological dynamics, not mathematical realities, inspire this perception. Statistically, the streak concept is nonsense. Humans also tend to subscribe to something researchers call “the law of small numbers,” which is the assumption that small samples faithfully represent entire populations. No scientific principle, however, underlies or enforces this “law.”

### Technical Description

Two primary interpretations of *representativeness bias* apply to individual investors.

1. *Base-Rate Neglect.* In base-rate neglect, investors attempt to determine the potential success of, say, an investment in Company A by contextualizing the venture in a familiar, easy-to-understand classification scheme. Such an investor might categorize Company A as a “value stock” and draw conclusions about the risks and rewards that follow from that categorization. This reasoning, however, ignores other unrelated variables that could substantially impact the success of the investment. Investors often embark on this erroneous path because it looks like an alternative to the diligent research actually required when evaluating an investment. To summarize this characterization, some investors tend to rely on *stereotypes* when making investment decisions.
2. *Sample-Size Neglect.* In sample-size neglect, investors, when judging the likelihood of a particular investment outcome, often fail to accurately consider the sample size of the data on which they base their judgments. They incorrectly assume that small sample sizes are *representative* of populations (or “real” data). Some researchers call this phenomenon the “law of small numbers.” When people do not initially comprehend a phenomenon reflected in a series of data, they will quickly concoct assumptions about that phenomenon, relying on only a few of the available data points. Individuals prone to sample-size neglect are quick to treat properties reflected in such small samples as properties that accurately describe universal pools of data. The small sample that the individual has examined, however, may not be representative whatsoever of the data at large.